

Cloud Foundations Virtual Bootcamp

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Week 7: Final Capstone Project

Submission Date: 27 August 2026

Business Problem: A start-up company needs a reliable cloud environment to host its website

Project: Deploy a company website on AWS

The company website I created is called CloudNova Technologies

1. VPC

The screenshot shows the AWS VPC console with a green success banner at the top: "You successfully created vpc-06eb54a9332a945c8 / cloudnova-vpc". The left sidebar shows the navigation menu with "VPC" selected. The main content area displays the details for the VPC "vpc-06eb54a9332a945c8 / cloudnova-vpc".

Details	
VPC ID vpc-06eb54a9332a945c8	State Available
DNS resolution Enabled	Tenancy default
Main network ACL acl-0da343223095d502b	Default VPC No
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled
Encryption control ID -	Encryption control mode -
Block Public Access Off	DHCP option set dopt-0e30260c756104799
IPv4 CIDR 10.0.0.0/16	Route 53 Resolver DNS Firewall rule groups -
DNS hostnames Disabled	Main route table rtb-05da09f408a68a48a
IPv6 pool -	Owner ID 754248136085

The screenshot shows the AWS VPC console with a green success banner at the top: "You successfully created vpc-06eb54a9332a945c8 / cloudnova-vpc". The left sidebar shows the navigation menu with "VPC" selected. The main content area displays "Your VPCs (2)" with a table listing the VPCs.

Name	VPC ID	State	Encryption c...	Encryption control ...
-	vpc-054b50d96b7d5ed4d	Available	-	-
cloudnova-vpc	vpc-06eb54a9332a945c8	Available	-	-

2. Public Subnet

The screenshot shows the AWS VPC console with a green success banner at the top: "You have successfully created 1 subnet: subnet-016b4dd49ff37a689". The left sidebar shows the navigation menu with "VPC" selected. The main content area displays "Subnets (1)" with a table listing the subnets.

Name	Subnet ID	State	VPC	Block Publi
cloudnova-pub-subnet	subnet-016b4dd49ff37a689	Available	vpc-06eb54a9332a945c8 clou...	Off

3. Internet Gateway

VPC > Internet gateways > igw-05909963a89b0dfc3

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

aws

Search

[Alt+S]

Ask Amazon Q

Europe (Stockholm)

Bakiimiiii3 (754248136085)

Thona8

The following internet gateway was created: igw-05909963a89b0dfc3 - cloudnova-IGW. You can now attach to a VPC to enable the VPC to communicate with the internet.

Attach to a VPC

igw-05909963a89b0dfc3 / cloudnova-IGW

Actions

Details info

Internet gateway ID

igw-05909963a89b0dfc3

State

Detached

VPC ID

-

Owner

754248136085

Tags (1)

Manage tags

Search tags

Key

Value

Name

cloudnova-IGW

The screenshot displays the AWS Management Console interface for an Internet Gateway. At the top, a green notification banner indicates that the Internet Gateway 'igw-05909963a89b0dfc3' has been successfully attached to the VPC 'vpc-06eb54a9332a945c8'. Below this, the page title is 'igw-05909963a89b0dfc3 / cloudnova-IGW'. The 'Details' section provides information about the gateway, including its ID, state (Attached), VPC ID, and owner. The 'Tags' section shows a single tag with the key 'Name' and value 'cloudnova-IGW'.

4. Route Table

The screenshot shows the AWS Management Console interface. At the top, a green banner indicates that the route table 'rtb-0ff9a30e0bf3ad61f' was created successfully. Below this, the breadcrumb navigation shows 'VPC > Route tables > rtb-0ff9a30e0bf3ad61f'. The left-hand navigation pane is expanded to 'Virtual private cloud', with 'Route tables' selected. The main content area displays the details for the route table 'rtb-0ff9a30e0bf3ad61f / cloudnova-pub-RT'. The details section includes fields for 'Route table ID' (rtb-0ff9a30e0bf3ad61f), 'Main' (No), 'Explicit subnet associations' (empty), 'Edge associations' (empty), 'VPC' (vpc-06eb54a9332a945c8 | cloudnova-vpc), and 'Owner ID' (754248136085). Below the details, there are tabs for 'Routes', 'Subnet associations', 'Edge associations', 'Route propagation', 'Tags', and 'Related resources'. The 'Routes' tab is active, showing a table with one route. The table has columns for 'Destination', 'Target', 'Status', 'Propagated', and 'Route Origin'. The first row shows a destination of '0.0.0.0/0' and a target of 'cloudnova-igw'.

Route table rtb-0ff9a30e0bf3ad61f | cloudnova-pub-RT was created successfully.

rtb-0ff9a30e0bf3ad61f / cloudnova-pub-RT

Details [info](#)

Route table ID rtb-0ff9a30e0bf3ad61f	Main No	Explicit subnet associations -	Edge associations -
VPC vpc-06eb54a9332a945c8 cloudnova-vpc	Owner ID 754248136085		

Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

Routes (1) [Both](#) [Edit routes](#)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	cloudnova-igw	Complete		Learned

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VPC > Route tables > rtb-0ff9a30e0bf3ad61f

Route table rtb-0ff9a30e0bf3ad61f | cloudnova-pub-RT was created successfully.

Details Info

Route table ID rtb-0ff9a30e0bf3ad61f	Main No	Explicit subnet associations -	Edge associations -
VPC vpc-06eb54a9332a945c8 cloudnova-vpc	Owner ID 754248136085		

Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (1)

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table

5. Security Groups

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EC2 > Security Groups > sg-089feafb2492ec982

Security group (sg-089feafb2492ec982 | cloudnova-EC2-SG) was created successfully

sg-089feafb2492ec982 - cloudnova-EC2-SG

Details

Security group name cloudnova-EC2-SG	Security group ID sg-089feafb2492ec982	Description Security group for ClodNova website	VPC ID vpc-06eb54a9332a945c8
Owner 754248136085	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules Outbound rules Sharing VPC associations Related resources - new Tags

Inbound rules (2)

Search

Security group rule ID	IP version	Type	Protocol	Port range	Source
sgr-0ab1f4dec2242fc3b	IPv4	SSH	TCP	22	41.13.96.1
sgr-0b3272303161ae9d4	IPv4	HTTP	TCP	80	0.0.0.0

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EC2 > Security Groups > sg-089feafb2492ec982

Security group (sg-089feafb2492ec982 | cloudnova-EC2-SG) was created successfully

sg-089feafb2492ec982 - cloudnova-EC2-SG

Details

Owner 754248136085	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry
-----------------------	---	--

Inbound rules Outbound rules Sharing VPC associations Related resources - new Tags

Inbound rules (2)

Search

Security group rule ID	IP version	Type	Protocol	Port range	Source
sgr-0ab1f4dec2242fc3b	IPv4	SSH	TCP	22	41.13.96.1
sgr-0b3272303161ae9d4	IPv4	HTTP	TCP	80	0.0.0.0

6. Launch an EC2 instance

The screenshot shows the AWS Management Console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and the account name 'Bakiiiiiii23 (754248136085)' in the 'Europe (Stockholm)' region. The main content area is titled 'Launch an instance' and features a green success banner: 'Success Successfully initiated launch of instance (i-010a85580ae149d60)'. Below this, there's a 'Launch log' section and a 'Next Steps' section with a search bar. Four cards are displayed: 'Create billing usage alerts', 'Connect to your instance', 'Connect an RDS database', and 'Create EBS snapshot policy'. The bottom screenshot shows the 'Instances' page with a table listing the launched instance 'cloudnova-EC2' with ID 'i-010a85580ae149d60', state 'Running', type 't3.micro', and availability zone 'eu-north-1a'. The instance details for 'i-010a85580ae149d60 (cloudnova-EC2)' are shown below the table, including the public IPv4 address '13.60.16.98' and private IPv4 addresses '10.0.1.32'.

Success
Successfully initiated launch of instance (i-010a85580ae149d60)

► **Launch log**

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
[Create billing alerts](#)

Connect to your instance
Once your instance is running, log into it from your local computer.
[Connect to instance](#)
[Learn more](#)

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
[Connect an RDS database](#)

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots.
[Create EBS snapshot policy](#)

Instances (1/1) Info
Last updated less than a minute ago
[Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets
Choose filter set
Find Instance by attribute or tag (case-sensitive)

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public
☒	cloudnova-EC2	i-010a85580ae149d60	Running	t3.micro	3/3 checks passed	eu-north-1a	-

i-010a85580ae149d60 (cloudnova-EC2)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

Instance summary Info

Instance ID i-010a85580ae149d60	Public IPv4 address 13.60.16.98 open address	Private IPv4 addresses 10.0.1.32
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7. Install and host a simple website

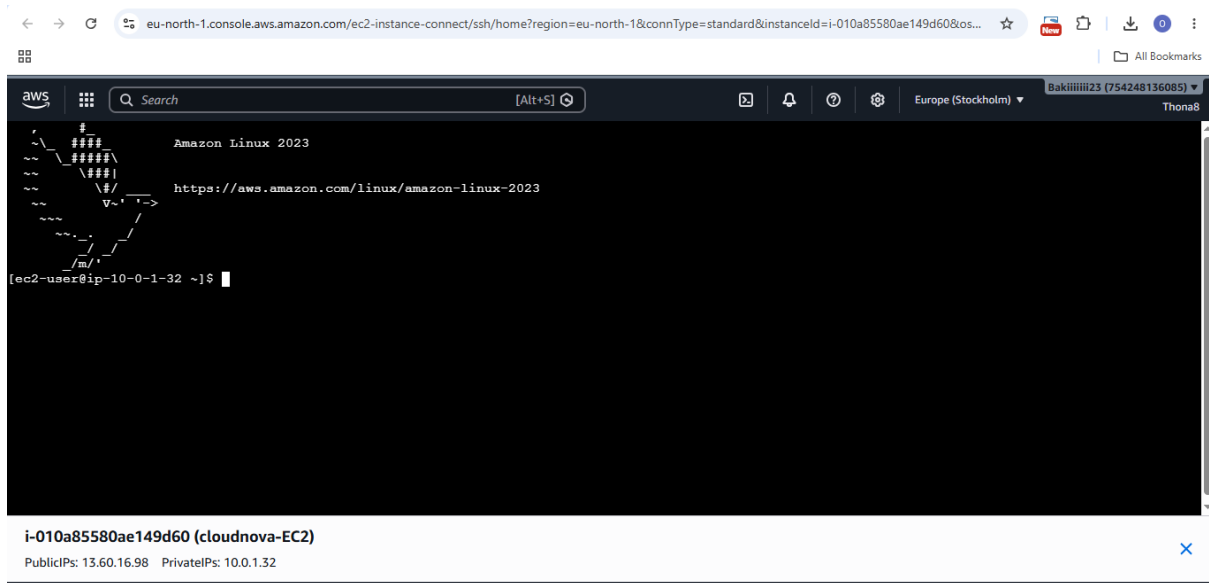
Before hosting my website, I checked if my IP address was live.

The screenshot shows a web browser address bar with the URL '13.60.16.98'. The browser indicates it's 'Not secure'. Below the address bar, there's a message 'It works!'.

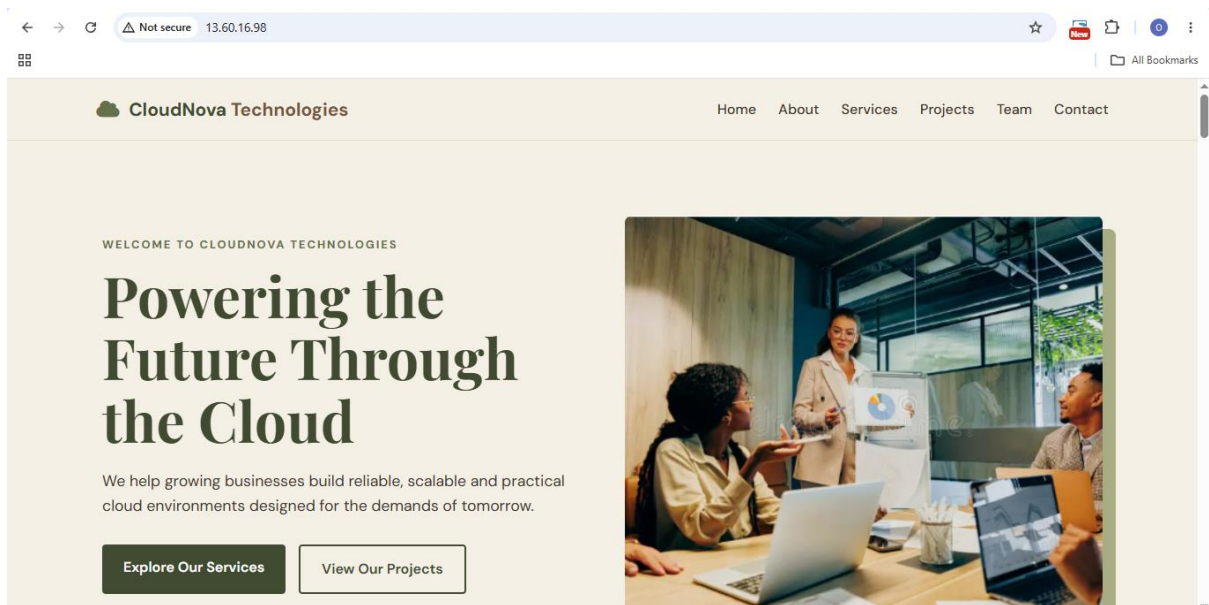
← → ↻ ⚠ Not secure 13.60.16.98

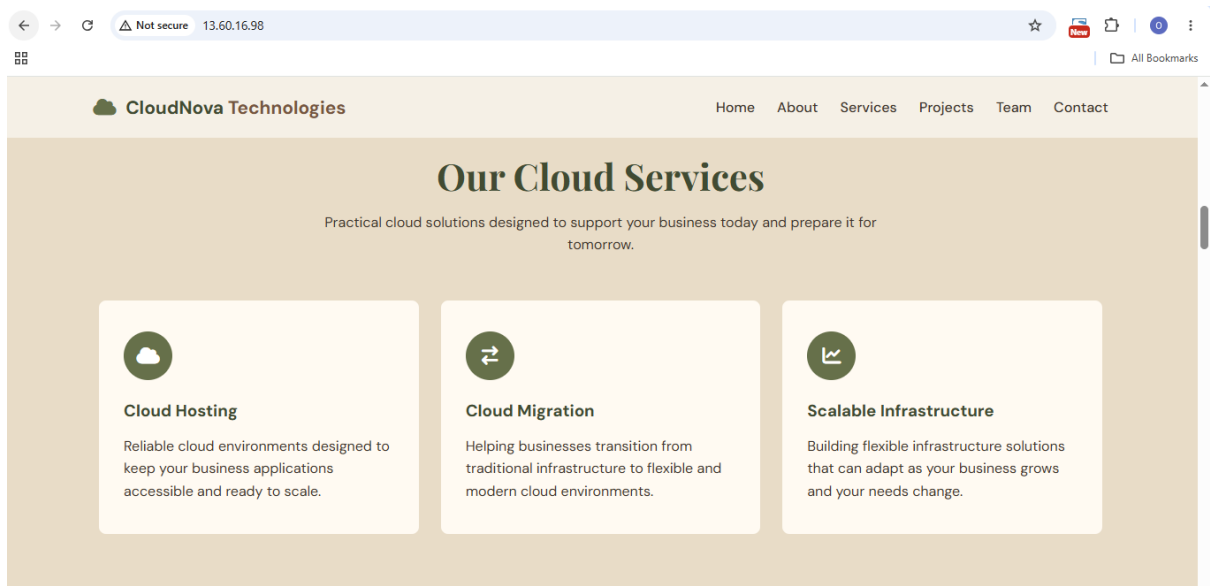
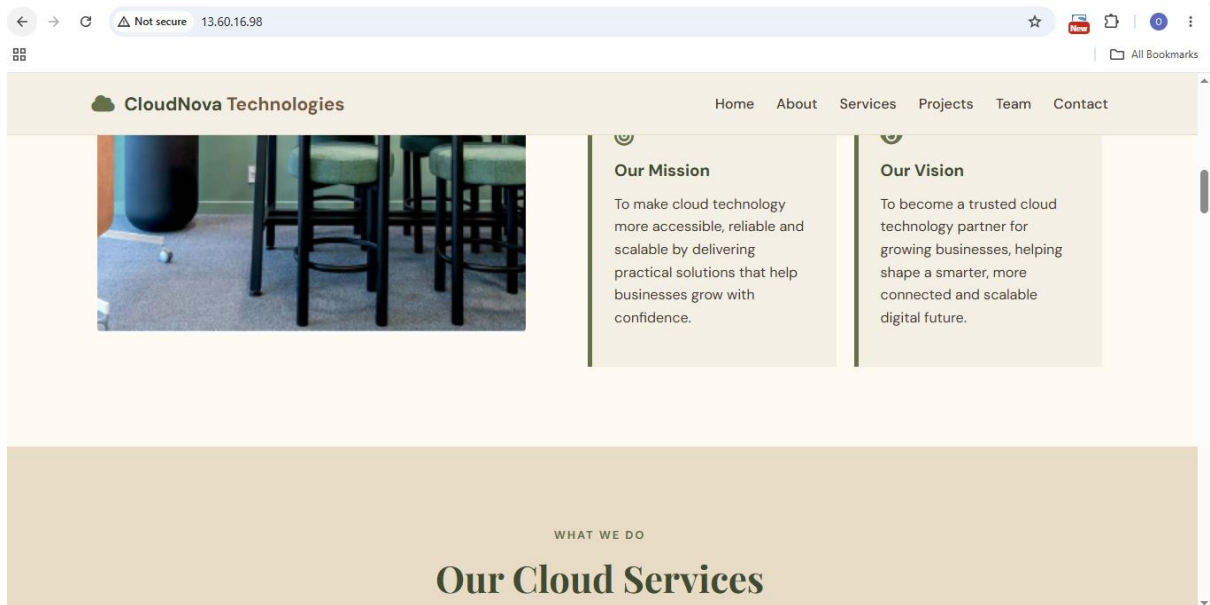
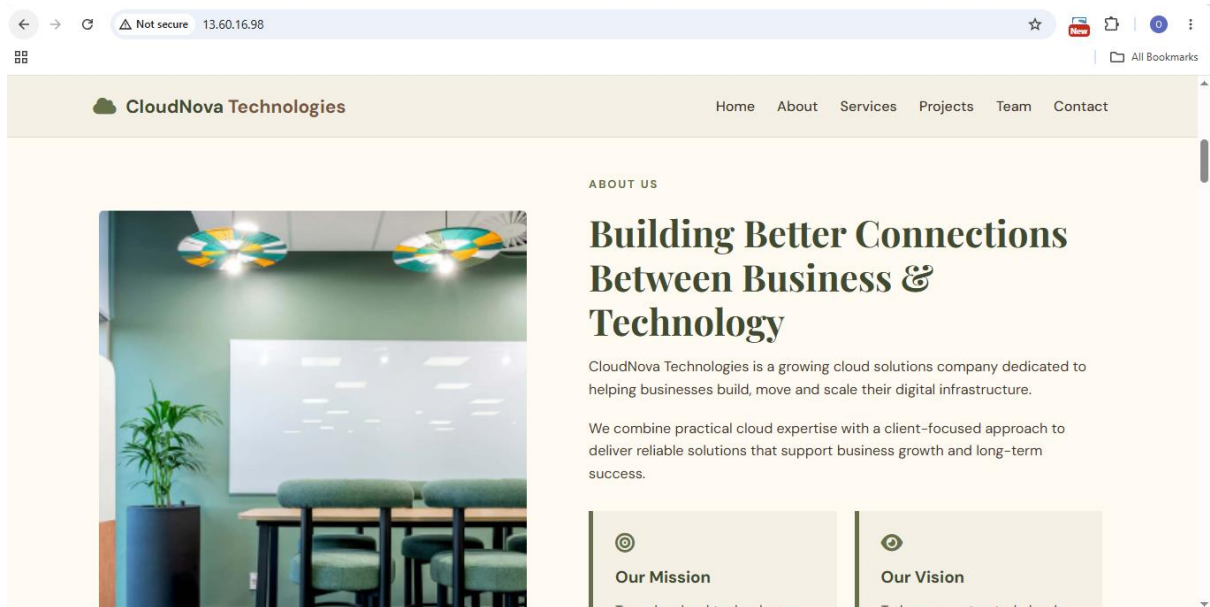
It works!

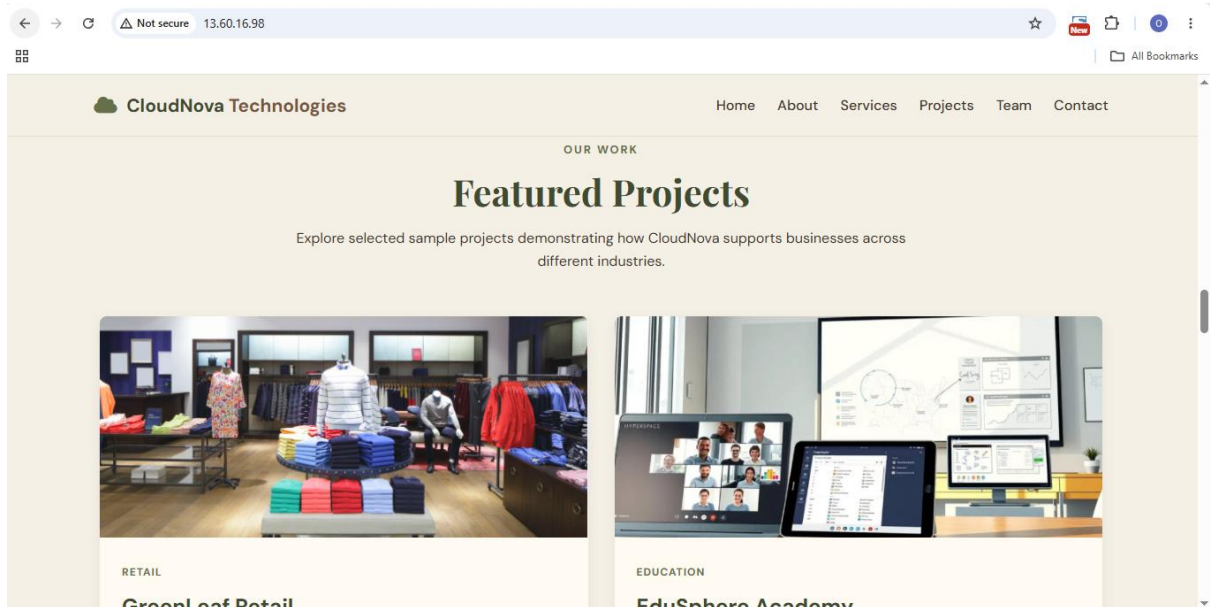
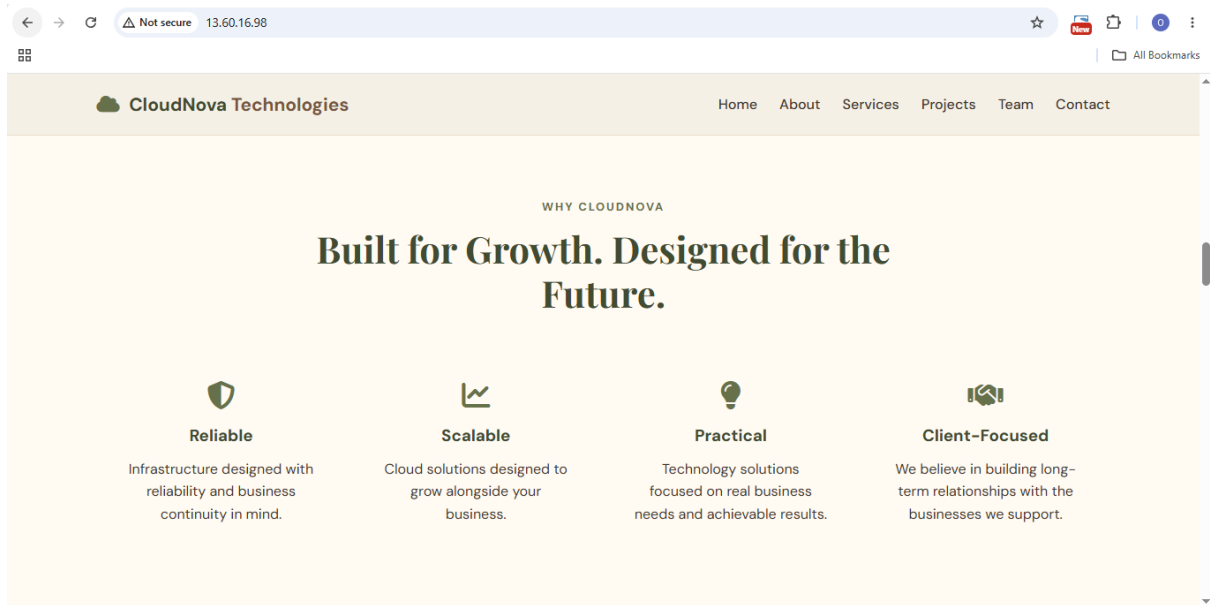
Installed web server on EC2 to put my static website files on to the server.



CloudNova website is hosted and live.







← → ↻ ⚠ Not secure 13.60.16.98

☆ 📄 📁 0 ☰

☰ All Bookmarks

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LOGISTICS

SwiftLogix

Developed a flexible infrastructure design to support growing operations and changing business demands.

Service: Scalable Infrastructure

View Case Study



HEALTHCARE TECHNOLOGY

MediConnect SA

Designed a secure and reliable cloud infrastructure approach to support the company's digital services.

Service: Cloud Infrastructure

View Case Study

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☰ All Bookmarks

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OUR CLIENTS

Trusted by Growing Businesses

GreenLeaf Retail

EduSphere Academy

SwiftLogix

MediConnect SA

OUR PEOPLE

← → ↻ ⚠ Not secure 13.60.16.98

☆ 📄 📁 0 ☰

☰ All Bookmarks

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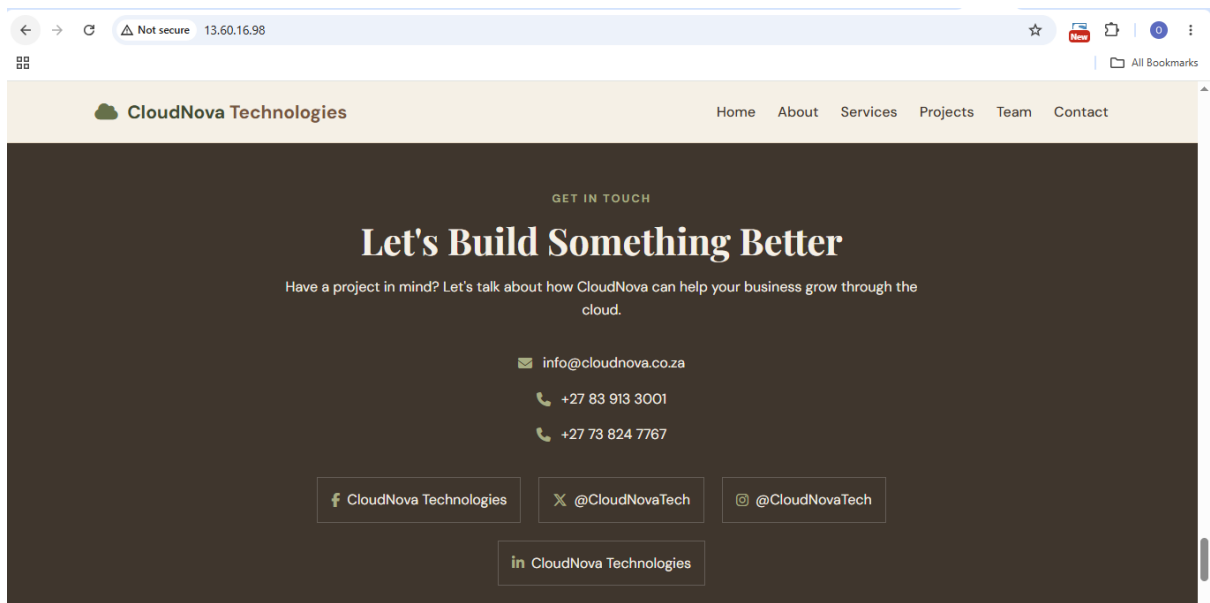
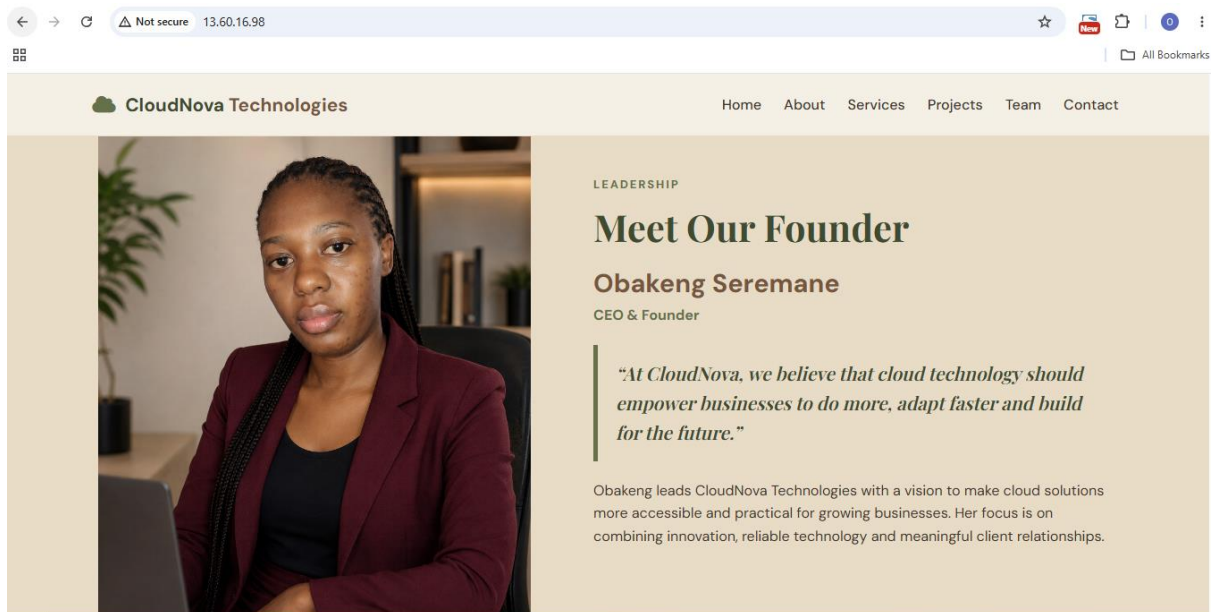
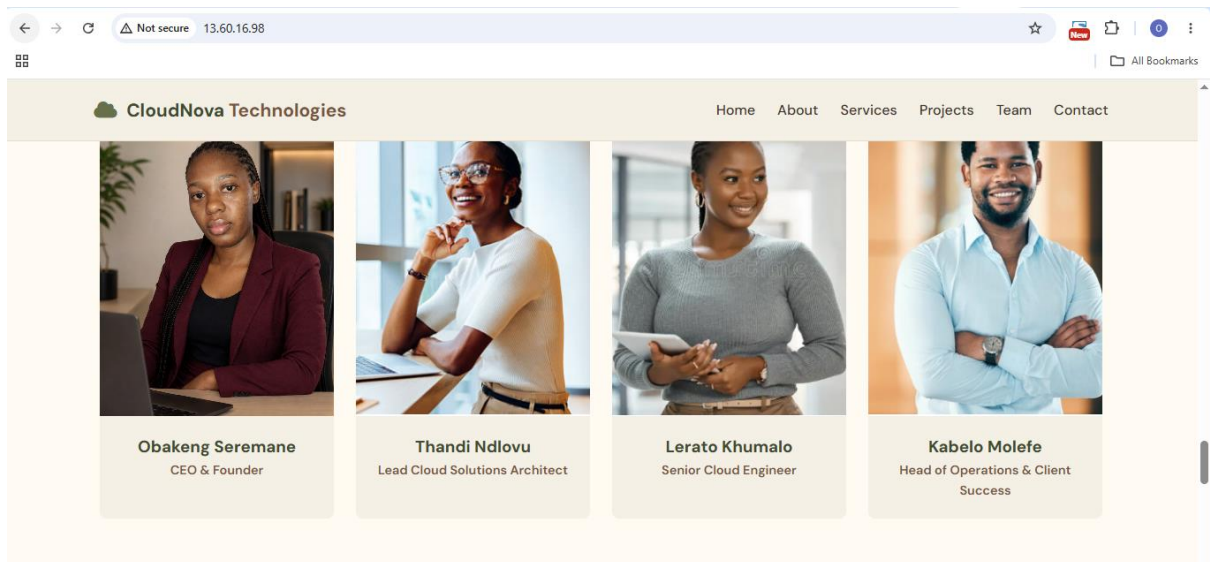
Home About Services Projects Team Contact

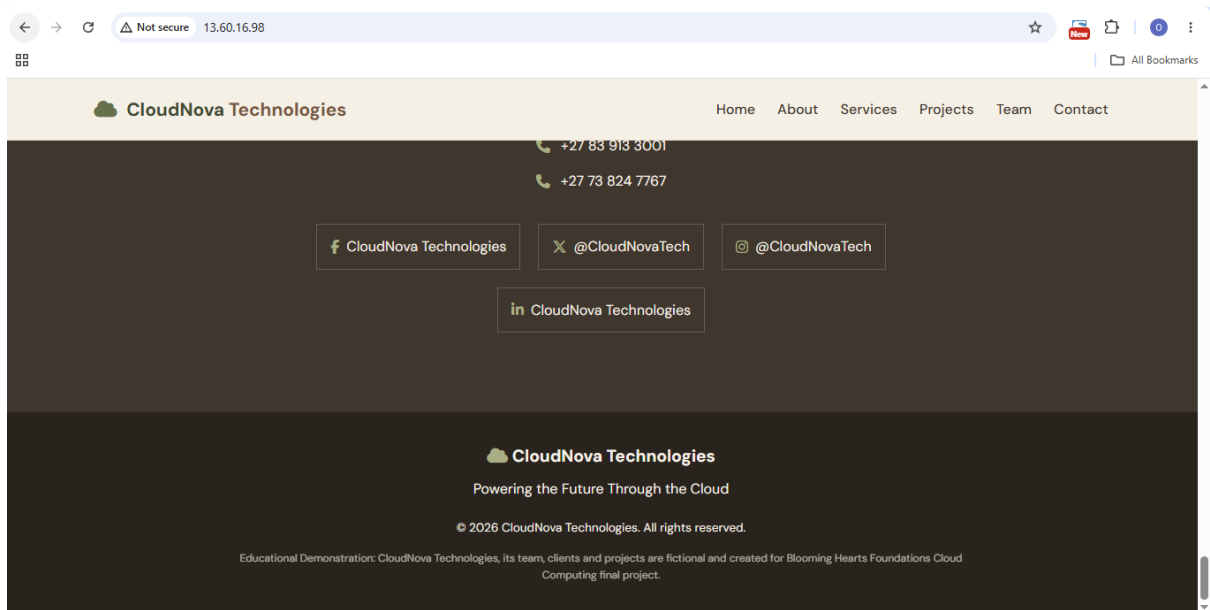
OUR PEOPLE

Meet the Team

Meet the people behind CloudNova Technologies.

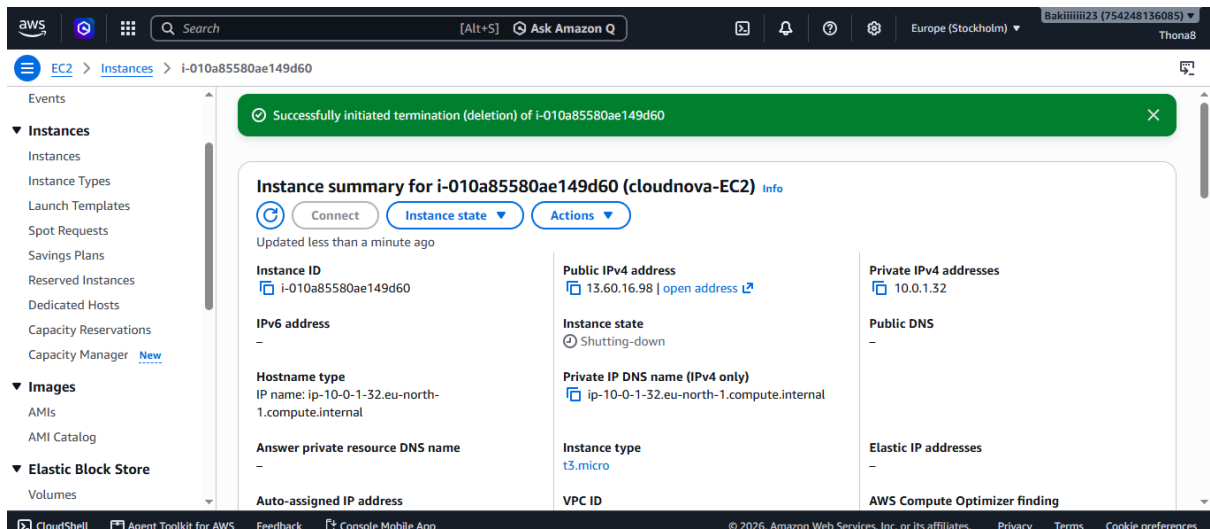






Resource Cleanup

1. Terminate and delete EC2



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EC2 > Instances

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager **New**

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Instances (1/1) Info

Last updated about 3 hours ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public
☒	cloudnova-EC2	i-010a85580ae149d60	Terminated	t3.micro	-	eu-north-1a	-

i-010a85580ae149d60 (cloudnova-EC2)

Details Status and alarms Monitoring Security Networking Storage Tags

▼ Instance summary Info

Instance ID

i-010a85580ae149d60

Public IPv4 address

-

Private IPv4 addresses

-

2. Delete Security Group

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EC2 > Security Groups

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager **New**

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IPs

Security group (sg-089feafb2492ec982 | cloudnova-EC2-SG) successfully deleted

Security Groups (3) Info

Export security groups to CSV

Create security group

Find security groups by attribute or tag

☐	Name	Security group ID	Security group name	VPC ID
☐	-	sg-09cd786789eb0ddbb	default	vpc-06eb54a9332a945c8
☐	-	sg-089feafb2492ec982	cloudnova-EC2-SG	vpc-06eb54a9332a945c8
☐	-	sg-01a0fe2806b0c0fd9	default	vpc-054b50d96b7d5ed4d

Select a security group

3. Delete Internet Gateway

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VPC > Internet gateways

Filter by VPC

AWS Global View

▼ Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Internet gateway successfully deleted - igw-05909963a89b0dfc3

Internet gateways (1) Info

Find internet gateways by attribute or tag

☐	Name	Internet gateway ID	State	VPC ID
☐	-	igw-05ef10e41436182a4	Attached	vpc-054b50d96b7d5ed4d

Select an internet gateway above

4. Delete Public Subnet

The screenshot shows the AWS Management Console interface for the 'Subnets' page. A green notification banner at the top states: 'You have successfully deleted subnet-016b4dd49ff37a689'. Below this, the 'Subnets (3)' section displays a table of subnets. The table has columns for Name, Subnet ID, State, VPC, and Block Public Access. Three subnets are listed, all with a state of 'Available'. The first two subnets are associated with VPC 'vpc-054b50d96b7d5ed4d', and the third is associated with 'vpc-054b50d96b7d5ed4d'. The 'Block Public Access' column shows 'Off' for all subnets. The left sidebar shows the 'Virtual private cloud' section with 'Subnets' selected. The top navigation bar shows the user is in the 'Europe (Stockholm)' region.

Name	Subnet ID	State	VPC	Block Public Access
-	subnet-095ee1e31265c97c3	Available	vpc-054b50d96b7d5ed4d	Off
-	subnet-09094a8de93b429b3	Available	vpc-054b50d96b7d5ed4d	Off
-	subnet-04aa8524928bf7875	Available	vpc-054b50d96b7d5ed4d	Off

5. Delete Route Table

The screenshot shows the AWS Management Console interface for the 'Route tables' page. A green notification banner at the top states: 'You have successfully deleted rtb-0ff9a30e0bf3ad61f / cloudnova-pub-RT'. Below this, the 'Route tables (2)' section displays a table of route tables. The table has columns for Name, Route table ID, Explicit subnet associations, Edge associations, Main, and VPC. Two route tables are listed, both with a state of 'Available'. The first route table is associated with VPC 'vpc-054b50d96b7d5ed4d', and the second is associated with 'vpc-054b50d96b7d5ed4d'. The 'Main' column shows 'Yes' for both route tables. The left sidebar shows the 'Virtual private cloud' section with 'Route tables' selected. The top navigation bar shows the user is in the 'Europe (Stockholm)' region.

Name	Route table ID	Explicit subnet associations	Edge associations	Main	VPC
-	rtb-05da09f408a68a48a	-	-	Yes	vpc-054b50d96b7d5ed4d
-	rtb-0df95e0ecd6b8ba0	-	-	Yes	vpc-054b50d96b7d5ed4d

6. Delete VPC

The screenshot shows the AWS Management Console interface for the 'Your VPCs' page. A green notification banner at the top states: 'You successfully deleted vpc-06eb54a9332a945c8 / cloudnova-vpc'. Below this, the 'Your VPCs (1)' section displays a table of VPCs. The table has columns for Name, VPC ID, State, Encryption configuration, and Encryption control. One VPC is listed, with a state of 'Available'. The VPC is associated with 'vpc-054b50d96b7d5ed4d'. The left sidebar shows the 'Virtual private cloud' section with 'Your VPCs' selected. The top navigation bar shows the user is in the 'Europe (Stockholm)' region.

Name	VPC ID	State	Encryption configuration	Encryption control
-	vpc-054b50d96b7d5ed4d	Available	-	-